Last fall degree lower bound: attempt and obstructions

A growing lower bound on the last fall degree of the binary Weil descent of Sem3: attempt and obstructions

Date: 2026-11-11. Treatise open problem 12 (`papers/ecdlp-index-calculus-treatise/sections/prog_openproblems.tex`). Status: research note, committed with #3878 and updated through the C runs of #3887, #3891 and #3893. Nothing was run off this machine (a 4-vCPU, 15 GB cloud container). No `pcc` claim exists for anything below, so every statement is `[pcc:unverified]`. The labels **PROVED**, **SKETCH**, **MEASURED**, **HEURISTIC** and **CONJECTURE** below say what kind of support each statement has.

1. Question

Let `F = F(n, V, z)` be the Boolean Weil descent of `Sem3(X1, X2, z)` over `K = GF(2^n)`, with `X1, X2` ranging over an `F2`-subspace `V` of dimension `k` (`N = 2k` Boolean variables, field equations implicit). For unsatisfiable `F`, the last fall degree `d_F` is the least `D` with `1 in W_D(F)` (treatise `falldegree.tex`, Prop. `bridges`). The question has two parts:

- Is there a lower bound on `d_F` that provably grows with `n` (or `k`), for any presentation of the system?
- If not, what precisely prevents a proof?

Short answer. No growing lower bound is proved here for `d_F`, in any presentation. What this note establishes:

1. **The literal "any presentation" question has a negative answer** (Lemma 2, PROVED). Every unsatisfiable instance has a degree-2 presentation of the same solution set with `d_F = 2`. It uses `sum_{j=2..D} C(N, j)` extension variables and *instance-independent* quadratic axioms. So a meaningful lower bound must either fix the presentation or bound its size.
2. **Fixing the presentation up to every natural re-encoding gives one number** (Lemma 1, PROVED). The re-encodings are a basis of `K`, a basis of `V`, an affine change of variables, and a basis of the equation span.
3. **Three proof routes have barriers:**
   - restriction (Lemma 4 ceiling, PROVED);
   - generator degree (Lemma 3, PROVED; it is the mechanism of every growing bound proved so far in the repository);
   - locality and expansion (Lemma 5, PROVED invariant, MEASURED size).
4. **A weaker, well-defined proof system has a measured linear growth.** This is the "selector" Nullstellensatz, the proof system of the treatise's `k+2` upper bound.
   - Its degree has an exact dual characterisation (Prop. S, S*, PROVED).
   - Its degree was measured at `ceil(k/3) + 2` on the diagonal for `k = 4..12`, which gave Conjecture S.

    - **Conjecture S is false as stated** (§6, COUNTEREXAMPLE, checked independently in both directions). At `n = 25`, `k = 13`, both instances tested have `e_min = 4 < ceil(13/3) = 5`. At `k = 14, 15` the measured value is back at `ceil(k/3)`. The selector degree still grows, but not as `ceil(k/3)`.
    - Local (flat-supported) dual certificates cannot prove more than `e >= 1` (Lemma F, PROVED, with a MEASURED premise).
    - Even a proof of Conjecture S would not bound `d_F`: measured `d_F` is 1–2 below the selector degree.

2. Sources checked

- **research-kb: not available.** `scripts/research-kb status` prints "research-kb: no executable found". No MCP server `research-kb` is configured in this session. So no `source_id` receipts exist. Repository files and the literature were read instead.
- **Sibling `isogeny-labs/crypto-autoresearcher`:** `knowledge/leads/ecdlp/*` (README.md, REVIEW_STATE.json, coverage.tsv) begin with the bytes `VLT1`. They are vault-encrypted and unreadable here; no decryption was attempted. The lead atlas (`tools/lead_atlas.py --brief`) was therefore not consulted.
- Repository (`/home/user/crypto`), all read in this session:
    - `papers/ecdlp-index-calculus-treatise/sections/falldegree.tex` (whole chapter);
    - `.../sections/prog_openproblems.tex` (problems 1, 12, 13);
    - `.../sections/prog_theory.tex`;
    - `.../ecdlp-index-calculus-treatise` bibliography entries `hky2015`, `kosters2015notes`, `kousidis2019`, `caminata2023`, `cei1996`, `iterated2023`;
    - `research/guarded_fall_degree_20261028/{README.md, STATUS.md, extensions/growing_b/proof.txt}` (formal/ listed, not re-read);
    - `research/prime_fall_degree_20261027/DERIVED.md` (D1–D8);
    - `research/pkm_falldegree_unblock_20261107/README.md` (launch logistics only, no theory);
    - `research/fall_degree_bounds_20261008/RESULTS.md` (E1–E8 headline);
    - `research/notes/index-calculus/RESEARCH_FALL_DEGREE_BOUNDS.md` (§0–2.4, references);
    - `research/notes/index-calculus/RESEARCH_FFD_PROOF_COMPLEXITY.md` (§1–3).
- Literature fetched in this session (WebSearch / WebFetch):
    - HKY, ePrint 2015/573 (abstract);
    - Kosters–Yeo, arXiv:1503.08001 (abstract);
    - Kousidis–Wiemers, arXiv:1906.05594 (abstract);
    - Caminata–Gorla, arXiv:2112.05579 (abstract);
    - Huang, arXiv:2103.07282 (abstract);
    - Gorla–Müller–Petit, arXiv:2011.01050 (ar5iv full text, summarised by the fetch tool);
    - Steiner, arXiv:2310.03637 (abstract).
- Cited from the repository ledger or from memory, **not re-read here** (marked † in the table):
    - Petit–Quisquater 2012;
    - Hodges–Petit–Schlather 2014;
    - Semaev–Tenti 2021;
    - HKYY arXiv:1505.02532;
    - Ding–Schmidt 2013;
    - Dubois–Gama 2010;
    - Ding–Hodges 2011;

- Bardet–Faugère–Salvy;
- the proof-complexity papers: Razborov 1998, Ben-Sasson–Impagliazzo 1999, Alekhnovich–Razborov 2001/2003, Buss–Grigoriev–Impagliazzo–Pitassi 2001, Clegg–Edmonds–Impagliazzo 1996.

3. Literature status

“Presentation” says whether the statement holds for one fixed generating set or for every one. Quotes are from the fetched abstract or text. Entries marked † are cited second-hand and need checking against the paper before they are quoted.

result	**quantity, direction**	**presentation**	**field**	**statement (source)**
Petit–Quisquater, ASIACRYPT 2012†	first fall, **upper** `<= m^2+1`	the Weil descent of `S_{m+1}`	`q = 2` (char 2)	Prop. 1 (ledger §2.1). The subexponential claim *assumes* `d_reg ≈ d_ff`.
Hodges–Petit–Schlather, FFA 30 (2014)†	first fall, **upper** (2-weight bound)	any Weil descent	any `q`	ledger §2.1
Kosters–Yeo, arXiv:1503.08001	first fall `= 2` for `Sem3` (a relation, “usually” a fall)	descent of `Sem3`	char 2, ordinary curves	abstract: first fall degree 2, “much lower than expected”; NP-completeness of “check if `S_r` evaluates to zero on some input” is conditional (the abstract does not name the assumption)
Kousidis–Wiemers, JMC 2019	first fall, **upper** `<= m^2 - m + 1` (ledger Thm 3.2)	descent of `S_{m+1}`	binary	abstract: “We improve on the first fall degree bound of polynomial systems that arise from a Weil descent”
HKY, CRYPTO 2015 / ePrint 2015/573	last fall: defined; **upper** for HFE	per presentation (the definition depends on it)	any `q`	abstract: “rigorously bounding last fall degrees of Weil descent systems remains an interesting but challenging open problem”
HKYY, arXiv:1505.02532†	last fall, **upper**, `n`-independent in `q, m, d_F(F), deg F, #solutions`	descent of a zero-dimensional radical `F`	any `q`	Thm 1.1 (ledger §2.3). Applied to `{S3, L_V(X1), L_V(X2)}` it gives `<= 4k+6`, linear in `n`.

result	quantity, direction	presentation	field	statement (source)
Huang, arXiv:2103.07282	last fall, **upper**	descents of linearised `F`	any `q`	abstract: relates the last fall degrees of `F_1` and `F'_1`, “As an application we derive upper bounds…”
Gorla–Müller–Petit, JSC 109 (2022), arXiv:2011.01050	solving degree and last fall, **upper** for HFE (Thm 5, Thm 11, Cor. 1 as summarised); **no lower bound**	HFE descent with field equations	any `q`	text: “further work needs to be done to close the gap between experiments and rigorous bounds”; Example 5: the Thm 11 bound “can be reached” in one case
Caminata–Gorla, JSC 2023, arXiv:2112.05579	`sd = max(d_F, max GB degree)` for degree-compatible orders; arbitrary gaps between first fall and `d_F`/`sd`/`d_reg` (Ex. 4.1–4.4, ledger)	one presentation each	any	abstract: “a connection between the solving degree and the last fall degree”
Semaev–Tenti, J. Algebra 2021†	`sd <= 2 d_reg - 2` with field equations	one presentation	any `q`	ledger §1
Ding–Schmidt 2013†	the degree of regularity can differ from the solving degree	examples	finite fields	not re-read
Dubois–Gama 2010; Ding–Hodges 2011†	HFE `d_reg`, **upper**	HFE	any `q`	not re-read
Bardet–Faugère–Salvy†	`d_reg` of semi-regular sequences	generic systems	any	applies only *if* semi-regular; not proved for these systems
Steiner, arXiv:2310.03637	**lower** bounds on Castelnuovo–Mumford regularity (MiMC family), not on the solving degree	iterated systems	prime fields	abstract: lower bounds show regularity-based solving-degree estimates “can never surpass a certain threshold”; they are not `sd` lower bounds
Razborov 1998; Ben-Sasson–Impagliazzo 1999; Alekhnovich–Razborov 2001/03; BGIP 2001†	Polynomial Calculus degree, **lower** `Omega(n)`	fixed local encodings: PHP, random k-CNF, expanding constant-arity systems	any field (BSI: char ≠ 2)	only for systems whose generators each depend on `O(1)` variables with an expanding

result	quantity, direction	presentation	field	statement (source)
				dependency graph. By Clegg–Edmonds–Impagliazzo, PC degree = d_F for unsatisfiable Boolean systems (ledger §1)
repo: guarded chain (growing_b/proof.txt)	d_F >= max(5, B+1), d_ff = B+1	explicit membership g(x) of degree B	F_p, p > 3; binary variant	generator-degree mechanism (Lemma 3 below)
repo: PKM naive control D5 (DERIVED.md)	d_F >= 2^t	{S3, x_i^K - g^K}	F_p	generator-degree mechanism
repo: treatise Prop. lower3, Thm. lfd5, lfdexact	d_F >= 3 (genericity conditions); >= 5 certified on named instances; R35a exactly 5	the Boolean descent	q = 2	constants and instances, not growth

Status. In published work every proved bound on the last fall degree or the solving degree of a Weil-descent system is an upper bound. Every proved lower bound that grows comes from the degree of an explicit generator (degree-B membership, x^{2^t}), or from proof complexity for *local* encodings. None applies to the fixed-degree Boolean descent of Sem3. This agrees with the open-problem statements of HKY 2015 and Gorla–Müller–Petit quoted above, and with Kousidis–Wiemers as cited in the treatise.

4. Setting and conventions

- **Closure.** B = F2[x_1..x_N]/(x_i^2 + x_i), with degree measured on multilinear representatives. W_D(F) is the smallest subspace of B_{<=D} that contains F ∩ B_{<=D} and contains x_i g whenever g in W_D(F) and deg g <= D-1. This is the convention of the ledger (§0) and of Prop. bridges, and equals the image of the HKY closure of F plus the field equations.

- **Degrees.** d_F is the least D with 1 in W_D. M_D is the span of the m f with deg m + deg f <= D, and d_NS is the least D with 1 in M_D.

- **The descent in (A, C) form.** Write X1 = A, X2 = A + C, so C = X1 + X2; A and C range over V, with coordinate vectors a, c. This is an invertible linear change of variables. With S = X1^2 X2^2 + z^2 (X1^2 + X2^2) + z X1 X2 + a6:

  ```
  S(A, C) = Phi*_C(A) + h_C,
  Phi*_C(A) = A^4 + (C^2 + z) A^2 + z C A,        (F2-linear in A for fixed C)
  h_C       = z^2 C^2 + a6                        (F2-affine in C).
  ```

  Each Boolean coordinate f_i = Tr(theta_i S) therefore has the form a^T M_i c + (linear in a) + (affine in c). It is affine in a for fixed c and affine in c for fixed a, as the treatise's proof of Thm. upper uses. The decomposition at C is unsatisfiable iff h_C ∉ Phi*_C(V).

5. What is proved here

Lemma 1 (orbit invariance). PROVED

Let `F, F'` be finite subsets of `B_{<=D}` with `span F = span F'`. Then `W_D(F) = W_D(F')`. If `phi` is an affine automorphism of `F2^N`, then `phi*(W_D(F)) = W_D(phi* F)`.

Proof. `W_D(F)` is a subspace containing `F`, so it contains `F'`. It is closed under the rule, so it contains `W_D(F')`; by symmetry the two are equal. Affine substitution and its inverse do not raise multilinear degree, so `phi*` maps the closure rule onto itself. ∎

Consequence: `d_F` is the same for every choice of trace basis of `K`, basis of `V`, affine coordinates, and basis of the equation span, once `D >= 2`. Within this orbit, “for any presentation” is the same as “for this presentation”.

Lemma 2 (extension variables collapse `d_F` to 2). PROVED; checked computationally

Let `F ⊂ B_N` be unsatisfiable with `max deg F <= 2 <= D := d_F(F)`. - **Extension variables.** For every monomial `m` with `2 <= |m| <= D`, introduce a variable `y_m`. Write `Y(∅) = 1`, `Y({i}) = x_i`, and `Y(m) = y_m` otherwise. - **Axioms.** `E_D = { x_i Y(m) + Y(m ∪ {i}) : |m| <= D-1, i in [N] }`. The trivial ones (`|m| <= 1` and `i in m`) are dropped. - **Linearised generators.** `f^ = sum_m c_m Y(m)` for `f = sum_m c_m m`. - **The extended system.** `F^ = {f^ : f in F} ∪ E_D`.

Then: - (a) the Boolean zeros of `F^` are exactly the lifts `y_m = prod_{i in m} x_i` of zeros of `F`, so `F^` is unsatisfiable; - (b) `1 in W_2(F^)`, so `d_F(F^) <= 2`.

Proof. - (a) By induction on `|m|`, the axiom with `m' = m \ {i}` forces `Y(m) = x_i Y(m') = prod x`. Then `f^ = f(x)`. - (b) Let `L` be the linear map `m ↦ Y(m)` on `B_{N,<=D}`. The set `G = {g in B_{<=D} : L(g) in W_2(F^)}` is a subspace containing `F`, since `L(f) = f^`. Suppose `g in G` with `deg g <= D-1`. Then `L(x_i g) = x_i L(g) + sum_m c_m (x_i Y(m) + Y(m ∪ {i}))`. - The first term lies in `W_2(F^)`, because `deg L(g) <= 1`. - The sum is a combination of axioms of degree `<= 2`, each in `W_2(F^)`. The dropped trivial axioms are 0 in `B`.

So `G` is closed under the rule and contains `W_D(F)`, which contains 1. Since `L(1) = 1`, we get `1 in W_2(F^)`. ∎

Remarks: - Only the linear forms `f^` depend on the instance. `E_D` depends only on `(N, D)`. - With `D <= k+2` (Thm. `upper`), every instance of a given `(n, k)` has such a presentation, with `N + sum_{j=2..k+2} C(2k, j)` variables. - A “for any presentation” lower bound on `d_F` is therefore **false** as stated. The right question is a trade-off between degree and size (number of variables). This note claims no theorem about that trade-off.

Computational check. `ext_check.py` builds `F^` for unsatisfiable instances. It gives `1 ∉ W_1(F^)` and `1 in W_2(F^)` on all 6 instances:

instance	d_F	variables of F^
n = 7 (3 instances)	3	92
n = 9 (2 instances)	3	175
n = 11 (1 instance)	3	298

Output is in `out/ext.txt`.

Lemma 3 (generator-degree lemma). PROVED

Let `F = F_0 ∪ F_1`, where `F_0` has a common Boolean zero (in the polynomial-ring version: a common zero over the algebraic closure) and every element of `F_1` has degree `>= D_1`. Then `1 ∉ W_D(F)` for every `D < D_1`, so `d_F(F) >= D_1`.

Proof. For `D < D_1`, `F ∩ B_{<=D} = F_0 ∩ B_{<=D}`, so `W_D(F) = W_D(F_0)`. This lies in the ideal of `F_0`, which vanishes at the common zero, while 1 does not. ∎

This is the mechanism of every growing lower bound proved in the repository: - the guarded chain `d_F >= B+1` (`growing_b/proof.txt` §4 adds the "no hidden falls in the core" argument to reach `B+1`); - the PKM naive control D5, `d_F >= 2^t`.

It can never give more than `max deg F`. For the descent (`max deg F = 2`) it gives nothing. This is the precise reason those results "do not transfer" (`STATUS.md`: "Cardinality lower bounds do not transfer").

Lemma 4 (restriction, and its ceiling). PROVED

Let `phi: F2^{N'} -> F2^N` be affine and `phi*: B_N -> B_{N'}` the induced substitution. Then `phi*(W_D(F)) ⊆ W_D(phi* F)`, so `d_F(phi* F) <= d_F(F)` for unsatisfiable `F`.

Proof. `phi*` does not raise degree. Take `g` with `phi* g in W_D(phi* F)` and `deg g <= D-1`. Then `phi*(x_i g) = (beta + sum_j alpha_j x'_j) phi*(g)`, and each term is admitted by the rule. ∎ This is the Boolean form of `DERIVED.md` D6.

Corollaries: - **(4a) Monotone in `V`.** If `V' ⊆ V`, then `d_F(F_{V'}) <= d_F(F_V)`, for unsatisfiable `F_V`. - Checked: `dF_restricted_k-1 <= dF` on every instance that has the column (`out/n7-11.txt`, `out/n13.txt`, `out/n15.txt`, `out/n17.txt`). - **(4b) Monotone in `t` for Semaev's chain (5).** Substitute `X_t := v in V` and `Y_{t-2} := y*`, where `Sem3(y*, v, R_X) = 0`. This turns the last equation into the zero function and leaves exactly the `(t-1)`-chain with target `y*`. Hence `d_F(chain_t(R_X)) >= d_F(chain_{t-1}(y*))`. - **SKETCH:** written for the chain as the treatise states it; not checked computationally. - **(4c) Ceiling.** Restrict `c` to an affine flat of dimension `j` (or `a` to one). The restricted system has `d_F <= j + 2`. - Proof: it is affine in `a` for each value of the `j` flat coordinates, and the indicator construction of Thm. `upper` uses multipliers of degree `j`. - Fixing `c` entirely gives `d_F <= 1`. - Hence a lower bound obtained by restriction can exceed `j + 2` only if the restricted system keeps `>= d-2` free dimensions on both sides, i.e. it is again a full-size member of the family. Restriction transfers lower bounds; it cannot create them.

Lemma 5 (no local presentation in the orbit). PROVED invariant, MEASURED size

For quadratic `f in B`, let `B_f(x, y) = f(x+y) + f(x) + f(y) + f(0)`. This is an alternating bilinear form, and its rank is invariant under affine changes of variables. If `f` depends on only `r` variables, then `rank B_f <= r`.

For `f_delta = Tr(delta w)` with `delta in ker Tr` (treatise Thm. `reform`), the form is `[[0, M_delta], [M_delta^T, 0]]` with `M_delta[i][j] = Tr(delta v_i v_j / z)`, so its rank is `2 rank M_delta`. So every non-trivial generator in any coordinates of the orbit depends on at least `2 min_delta rank M_delta` variables.

Measured `min_delta rank M_delta` (`--rank`):

n	k	min rank
7	4	1–2
9	5	2
11	6	2–3
13	7	2–3
15	8	3–4

The rank histograms concentrate at `k-1` and `k` (`out/n15ns.txt`). A dimension count suggests the minimum is about `k - sqrt(2k)` (HEURISTIC).

Consequently the Alekhnovich–Razborov / Ben-Sasson–Impagliazzo expansion method has no `O(1)`-arity presentation to work on. This is a barrier to that method, not a proof that the method fails.

Remark (complexity). PROVED, elementary

Suppose `d_F <= c` held for all unsatisfiable `m = 2` instances, for a constant `c`. Then computing `W_c` (time `N^{O(c)}`) would decide decomposability in polynomial time. So any hardness result for the `m = 2` subspace-decomposition decision problem would imply that `d_F` is unbounded. None is known: the Kosters–Yeo NP-completeness is for `S_r` with growing `r`, and is conditional. Conversely, an unconditional `d_F -> infinity` faces no known complexity-theoretic barrier, since PC degree lower bounds are proved unconditionally for other families. The obstacle is technical.

6. The proof attempt: the selector proof system

The treatise's upper bound `d_F <= k+2` is a refutation of the form `1 = sum_i g_i(c) f_i`, with multipliers depending on `c = a + b` only. Call the least possible `max deg g_i` the **selector degree** `e_min`, and `d_SEL := least D with 1 in span{m(c) f : deg m + deg f <= D}`.

Trivially `d_F <= d_NS <= d_SEL <= k+2`, because the multiplier sets are nested. A lower bound on `e_min` is therefore the weakest statement in this chain. Even that is not proved; below are the exact reformulations and how far the proof attempt got.

Proposition S (field form of the selector system). PROVED; checked computationally

Write `lambda := sum_i g_i theta_i`, a map `V -> K` whose coordinate degree is `max deg g_i`. Then `sum_i g_i(c) f_i = 1` in `B` iff, for every `C in V`:

```
(I)  Tr(lambda(C) * Phi*_C(v)) = 0   for all v in V,
(II) Tr(lambda(C) * h_C)       = 1.
```

Proof. For fixed `c`, `sum_i g_i f_i = Tr(lambda(C) S(A, C)) = Tr(lambda(C) Phi*_C(A)) + Tr(lambda(C) h_C)`. This is affine in `A in V`. It is the constant 1 iff its linear part vanishes on `V` and its constant is 1. Identity of functions on `F2^{2k}` is identity in `B`. ∎

Check. `selector_check.py` solves (I)–(II) by linear algebra over `2^k` points. That code is independent of the ANF/Macaulay code that computes `d_SEL`. The two agree, `e_min + 2 = d_SEL`, on all 10 instances run with both:

n	instances
7	4
9	3
11	3

Proposition S* (exact dual: selector pseudo-solutions). PROVED

Let F be unsatisfiable and put Rc_C := Phi*_C(V) ⊕ F2 h_C. The sum is direct because h_C ∉ Phi*_C(V). No selector of degree <= e exists iff there is a map p: V -> K with: - (i) every coordinate of p of Boolean degree <= k-1-e. Equivalently, sum_{C in U} p(C) = 0 for every affine flat U of dimension k-e. - (ii) p(C) in Rc_C for every C. - (iii) #{C : p(C) ∉ Phi*_C(V)} is odd.

Proof. (I)–(II) is an affine system A lambda = b over F2 in the coefficients of lambda in P_e. It has no solution iff some y has yA = 0 and y.b = 1. - Write y = (y_C in F2^k, eps_C in F2)_C. Then y.(A lambda) = sum_C Tr(lambda(C) omega(C)) with omega(C) = Phi*_C(sum_j y_{C,j} v_j) + eps_C h_C in Rc_C. - In a trace-dual basis, yA = 0 says every coordinate of omega is orthogonal to RM(e, k), i.e. lies in RM(e, k)^perp = RM(k-e-1, k). - y.b = sum_C eps_C, and eps_C is the h_C-component of omega(C). - Take p = omega. ∎

For e >= k, (i) forces p = 0, so no dual certificate exists and a selector of degree k always exists. This is a second proof of Thm. upper.

Lemma F (flat-supported certificates are too weak). PROVED, with a MEASURED premise

Let U be an (e+1)-flat. Then p = rho * 1_U satisfies (i)–(iii) iff rho in cap_{C in U} Rc_C and #{C in U : rho ∉ Phi*_C(V)} is odd. These are exactly the obstructions visible after restricting to U. If cap_{C in U} Rc_C = 0 for every 2-flat U, flat restriction certifies nothing beyond e_min >= 1.

Measured (flat_check.py, 1,500 random flats per instance, out/flat_check.txt):

n	2-flat intersections that are 0	1-flat intersection dimension
7	1,345 / 1,409 and 1,275 / 1,405	mostly 1–2
9–13	≥ 98.9%	mostly 1–2
15	1,494 / 1,495 and 1,495 / 1,496	mostly 1–2

Rare 2-flats keep a 1-dimensional intersection, so they can only certify e >= 1. Any growing lower bound on e_min therefore needs genuinely *global* dual maps p of high Boolean degree (k-1-e) that satisfy moving pointwise constraints. The attempt stopped here.

Where the attempt broke. To prove e_min >= e_0, one needs to exhibit p in a space of dimension (k+1) 2^k (pointwise in Rc_C). It must satisfy n * sum_{j<=e_0} C(k, j) independent flat-sum conditions and have odd parity. Counting shows that many p exist with conditions (i)–(ii) for e_0 < k/2. Parity is a single linear functional, and showing it is not in the span of the conditions is the whole difficulty. That is primal infeasibility again; no structural construction of p was found.

Conjecture S (selector degree grows linearly). FALSIFIED as stated at k = 13

As first stated: for unsatisfiable random-subspace instances on the diagonal `k = ceil(n/2)`, `e_min >= ceil(k/3)`, so `d_SEL >= ceil(k/3) + 2`. The counterexample and the data beyond `k = 12` follow the original data below.

- **Data** (`out/emin_*.txt`, `out/n*.txt`):

k	4	5	6	7	8	9	10	11	12
e_min	2 (×5), 1 (×1)	2	2–3	3	3	3	4	4	4
ceil(k/3)	2	2	2	3	3	3	4	4	4

  The single exception is `n = 7` seed 16, an instance with `d_F = 2`. So the conjecture is stated for `k >= 5`.

- **Off-diagonal caveat.** At `n = 15` with `k = 7` (off the diagonal), `e_min = 2 < ceil(7/3)`. So `e_min` depends on `n` as well as `k`, and the conjecture is stated only for `k = ceil(n/2)`.

- Distinct instances behind the table: 6, 6, 6, 4, 4, 4, 3, 2, 2 for `k = 4..12`. They pool `d_SEL - 2` from `lfd_small.py` and `e_min` from `selector_check.py`; the two were shown equal at `n <= 11`.

- **Heuristic.** With `n ~ 2k` unknown coordinates times `C(k, <=e)`, against `(k+1) C(k, <=e+1)` constraints, solvability needs `e >~ (k-2)/3`.

Counterexample (`n = 25, k = 13`). COUNTEREXAMPLE, checked independently in both directions. `lfd_small.make_instance(25, seed)` for seeds 5 and 6 (the first two unsatisfiable seeds) gives `e_min = 4`, while `ceil(13/3) = 5`. - **`e_min <= 4` (feasibility).** `emin.c` found `lambda` with every coordinate of degree `<= 4` that satisfies all `2^13 * 14 = 114,688` constraints of Prop. S. - `verify_lambda.py` re-checks the saved `lambda` (`out/c/cert/n25_s{5,6}/lambda_e4.bin`) with `lfd_small`'s own field arithmetic: 0 violated constraints, ANF degree 4, for both seeds. - **`e_min > 3` (infeasibility).** The rows of 838 points (`out/c/cert/n25_s{5,6}/points_e3.txt`) are already inconsistent at `e = 3`. - `emin verify` rebuilds them by a second code path (`Tr(x^j w)` computed directly) with PLUQ elimination: rank 9,371 for the coefficient matrix, 9,372 with the right-hand side, for both seeds.

So Conjecture S fails as stated `[pcc:unverified]`. Proposition S, the definition of `e_min`, and the measurements at `k <= 12` are unaffected.

C data beyond `k = 12` (`emin.c`, `out/c/emin_n*.jsonl`; seeds are the first unsatisfiable ones):

n	k	seeds	e_min	ceil(k/3)	unknowns at the last e tested
23	12	4, 8	4, 4	4	18,262
25	13	5, 6	**4, 4**	5	27,325
27	14	3, 6	5, 5	5	93,771
29	15	3, 4	5, 5	5	143,376
31	16	2, 3	> 4, > 4	6	78,027
33	17	1, 5	> 4, > 4	6	106,062
35	18	6, 7	> 4, > 4	6	141,680
37	19	1, 3	> 3, > 3	7	42,920
39	20	3, 7	> 3, > 3	7	52,689

n	k	seeds	e_min	ceil(k/3)	unknowns at the last e tested
41	21	1, 2	> 3, > 3	7	64,042

- **Coverage.** `n = 15–23` reproduces the Python `e_min` on every committed instance, and every feasible `lambda` at `n = 25–29` passes `verify_lambda.py`.
- **Every lower bound at `n = 31–41` is independently re-checked.** `emin verify` (second row builder, PLUQ) confirms each point set is inconsistent; the right-hand side always raises the rank by one:
  - `n = 31`, `e = 4`: 77,451 → 77,452;
  - `n = 33`, `e = 4`: 105,365 → 105,366;
  - `n = 35`, `e = 4`: 140,846 → 140,847;
  - `n = 37`, `e = 3`: 42,748 → 42,749;
  - `n = 39`, `e = 3`: 52,498 → 52,499;
  - `n = 41`, `e = 3`: 63,831 → 63,832.

  The same ranks hold for both seeds at each `n`. Raw outputs are in `out/c/verify_*.txt`.
- **Above `n = 29` the rows are lower bounds only.** The next `e` needs `n * sum_{d<=e} C(k,d)` unknowns:
  - 213k (`n = 31`, `e = 5`), 310k (`n = 33`, `e = 5`), 442k (`n = 35`, `e = 5`), 186k (`n = 37`, `e = 4`), 242k (`n = 39`, `e = 4`), 309k (`n = 41`, `e = 4`).
  - The container restarts about hourly, which killed the 213k run twice. So `n = 31, 35, 37, 39` were finished with a 145k cap, keeping each test under about 20 minutes.
  - Exact values there need a host that stays up and has more memory, or a sparse solver.

What survives. The measured `e_min` is non-decreasing in `k` on this diagonal: 4, 4, 4, 5, 5 at `k = 11..15`. The `k = 13` value of 4 was certified on both instances. A weaker form, `e_min >= ceil((k-1)/3)`, fits every diagonal instance measured so far, including the `n = 7`, seed 16 exception (`k = 4`, `e_min = 1`). This form is a CONJECTURE fitted to the same data that falsified the original, so it is not evidence. The data at `k >= 16` cannot test it. Those rows are lower bounds of 4 or 3, below both `ceil(k/3)` and `ceil((k-1)/3)`, so they agree with either form and decide nothing.

Even a proof of Conjecture S would show only that the strategy of the `k+2` upper bound cannot be improved to constant degree with `c`-only multipliers. It says nothing about `d_F`. The measured `d_F` is smaller:

n	k	d_F	d_PCc	d_NS	d_SEL
7	4	3 (one instance 2)	3	4 (2)	4 (3)
9	5	3	3	4	4
11	6	3	3	4	4–5
13	7	3	**4**	4	5
15	8	4	4	5	5
17	9	4	4	5	5
19	10	4	4	5	6
21	11	4	—	—	6

`d_PCc` is the closure that may multiply by `c`-variables only. At `n = 13` (`d_PCc = 4 > d_F = 3`) the falls need multiplication by `a`-variables. These are exactly the falls of Lemma `deg3`,

`Tr(lambda X_i w)` for `lambda in V^perp`, which no selector can produce.


Last fall degree and selector degree against k on the diagonal, n <= 23 (figures/make_degrees.py, from the table above).
Last fall degree and selector degree against k on the diagonal, n <= 23 (figures/make_degrees.py, from the table above).

Why this does not extend to `d_F`

For `d_F` the dual object is a certificate in the sense of Prop. `cert`. It needs condition (ii), closure of the kernel under multiplication by variables, which is Razborov's pseudo-reduction condition. It is not linear in the certificate, so the clean Fredholm duality of Prop. S* is lost.

A 1-dimensional certificate is forced to be a genuine solution: - condition (ii) gives `pi(x u) = c_x pi(u)`, with `c_x^2 = c_x`, on `B_{<=d-1}`; - so certificates must have large rank, consistent with the certified codimensions 4,888–12,742.

No family of large-rank certificates with a uniform description was found. The closure data (`W_dims_at_dF_minus_1` at `n = 15–21`: `(0, 1, 3n, ...)`) reproduce Lemma `deg3`'s count `3n` for the degree-`<=2` part of `W_3`, and give no further handle.

7. Small computations

All exact over `F2`, pure Python 3.13 plus numpy, on this machine. Instances are seeded by `random.Random(seed)`, with: - `a2 in {0, 1}`; - `a6` random; - `z` the abscissa of a random curve point; - `V` a random `k`-dimensional subspace, `k = ceil(n/2)` unless `--k` is given; - the field `GF(2^n)` defined by the lowest-weight irreducible polynomial.

`--unsat m` scans seeds 1, 2, ... and keeps the first `m` unsatisfiable instances. The skipped satisfiable count is printed. Satisfiability is checked by exhaustive evaluation over all `2^{2k}` points (`lfd_small.py`) or by exact per-`C` linear algebra (`selector_check.count_solutions`); the two select identical seeds at `n = 11, 13`.

Commands, run from `research/lfd_lower_bound_20261111/`:

```
        for n in 7 9 11; do python3 lfd_small.py --n $n --unsat 6 --ns --sel --rank --
restrict --maxD 7; done > out/n7-11.txt
        python3 lfd_small.py --n 13 --unsat 4 --ns --sel --rank --restrict --maxD 7 >
out/n13.txt        # 16 s
        python3 lfd_small.py --n 15 --unsat 3 --sel --restrict --maxD 6 > out/n15.txt
# 5 s
        python3 lfd_small.py --n 15 --unsat 3 --ns --rank --maxD 6 > out/n15ns.txt
# 73 s
        python3 lfd_small.py --n 17 --unsat 3 --sel --restrict --maxD 6 > out/n17.txt
# 25 s
        python3 lfd_small.py --n 17 --unsat 2 --ns --maxD 6 > out/n17ns.txt
# 10 s
        python3 lfd_small.py --n 19 --unsat 2 --sel --maxD 6 > out/n19.txt
# 86 s
        python3 lfd_small.py --n 19 --unsat 1 --pcc --ns --maxD 6 > out/n19_pcc_ns.txt
# 52 s
        python3 lfd_small.py --n 21 --unsat 1 --sel --maxD 6 > out/n21.txt
```

```
# 173 s
        python3 lfd_small.py --n 15 --k 6 --unsat 3 --sel --ns --maxD 7 >
out/n15_kscan.txt              # also --k 7, --k 5
        for n in 7 9 11 13; do python3 lfd_small.py --n $n --unsat 4 --pcc --sel --maxD
7; done > out/pcc_7_13.txt
        for n in 15 17; do python3 lfd_small.py --n $n --unsat 3 --pcc --maxD 6; done >
out/pcc_15_17.txt
        python3 selector_check.py --n 7 --unsat 4; python3 selector_check.py --n 9 --
unsat 3; python3 selector_check.py --n 11 --unsat 3
        python3 selector_check.py --n 13 --unsat 4 --no-dsel
        for n in 15 17; do python3 selector_check.py --n $n --unsat 4 --no-dsel --maxe
7; done > out/emin_15_17.txt
        python3 selector_check.py --n 19 --unsat 3 --no-dsel --maxe 7 > out/emin_19.txt
# 33 s
        python3 selector_check.py --n 21 --unsat 2 --no-dsel --maxe 7 > out/emin_21.txt
# 71 s
        python3 selector_check.py --n 23 --unsat 2 --no-dsel --maxe 7 > out/emin_23.txt
# 229 s
        python3 ext_check.py --n 11 --unsat 1 > out/ext.txt; python3 ext_check.py --n 7
--unsat 3 >> out/ext.txt; python3 ext_check.py --n 9 --unsat 2 >> out/ext.txt
        for n in 7 9 11 13 15; do python3 flat_check.py --n $n --unsat 2 --samples 1500;
done > out/flat_check.txt
```

Raw outputs are in `out/`, one Python dict per instance, and include the curve, `z`, `V` and the field polynomial.

Agreement with the treatise. - `d_F = 3` at `n = 11, 13` and `d_F = 4` at `n = 15–21` on every instance. This matches Thm. `lfdexact` (3 at `n = 11, 13`; 4 for random subspaces at `15 <= n`). - `dim(W_3 ∩ B_{<=2}) = 3n` (45, 51, 57, 63 at `n = 15, 17, 19, 21`), as in Lemma `deg3`. - These are independent re-computations at sizes the treatise already covers. They are not new evidence about growth.

Off-diagonal (`n = 15`).

k	d_F	d_NS	d_SEL
5	2, 2	3, 3	3, 3
6	3, 3	3, 4	4, 4
7	3, 3, 3	4, 4, 4	4, 4, 4
8	4	5	5

At `k = 5` and `k = 6` a third seed (seed 3) is refuted at degree 1 by the linear trace equations alone.

8. Cheapest next step

1. **Done: `e_min` ported to C** (`emin.c`, #3887). It runs `n = 23` in 2 s, against 229 s for the Python version. Exact values hold to `n = 29`; above that, only independently checked lower bounds were obtained, up to `n = 41`.
    - The port falsified Conjecture S at `k = 13` (§6).
    - Exact values above `n = 29` need sparse or structured elimination, or a host with more memory (EKS). The dense system has 0.3–8M unknowns there.

    - Still open: solve the *dual* system of Prop. S* at `n = 13–25` and inspect the certificate `p` (support, degree, dependence on `C`). Comparing the `k = 13` instances with `k = 12, 14` should show what lets degree 4 suffice at `k = 13`.
2. **Measure `d_PCc` and `d_F` side by side at `n = 23–29`**, reusing the existing M4RI closure with a restricted multiplier set. This shows whether the `a`-multiplication falls (Lemma `deg3`) are the whole gap between `d_F` and `d_SEL`. If `d_PCc - d_F` stays at 0–1 while `d_SEL` grows, the next target is a pseudo-reduction that handles only those falls.
3. The guarded chain at `B = 4, 5` over a small `F_p` (5 variables, degree `<= 11`) is a cheap exact computation for problem 13. Lemma 3 shows its outcome cannot bear on problem 12.

9. Not claimed

- **No lower bound on `d_F`** that grows with `n`, `k`, `t` or `B` is claimed for the Boolean descent, in any presentation.
- **Lemma 2 does not claim** that small extended presentations exist; it uses `~N^{d_F}` variables. Nothing is claimed about size–degree trade-offs, Extended PC, or any proof-complexity simulation.
- **Lemma 5 does not prove** that expansion methods fail; it only shows that every presentation in the orbit has non-constant arity. The growth `k - sqrt(2k)` of the minimum rank is heuristic.
- **Conjecture S is false as stated** (`k = 13`, two instances). The weaker form `e_min >= ceil((k-1)/3)` is fitted to the same data and is not claimed. Any such bound would cover only the selector degree, never `d_F`. Propositions S and S* are exact reformulations, not bounds.
- **The `n = 31–41` rows are lower bounds** on `e_min` only. No exact value is claimed above `n = 29`.
- **Corollary 4b** (monotonicity in `t`) is a sketch for the chain as stated in the treatise; it was not checked computationally.
- **The small-size data are not evidence about asymptotic growth of `d_F`.** At `n <= 21` they reproduce known values (3, then 4).
- **No security, speed, ECDLP or subexponentiality claim** is made. Nothing here supports or refutes Semaev's or Petit–Quisquater's heuristics.
- **No `pcc` claim was filed.** All statements are `[pcc:unverified]`.
- Entries marked † in §3 were not re-read in this session.